

LATROBE VALLEY, Australia

WMO: 948910

Lat: 38.2094S

Lon: 146.4747E

Elev: 185

StdP: 14.60

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 32.1	(c) 34.4	(d) 30.8	(e) 25.1	(f) 35.5	(g) 32.7	(h) 27.4	(i) 38.3	(j) 28.7	(k) 50.0	(l) 25.6	(m) 51.3	(n) 1.5	(o) 230	(p) 0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 23.1	(c) 92.7	(d) 67.2	(e) 87.6	(f) 66.6	(g) 83.2	(h) 65.5	(i) 70.3	(j) 83.7	(k) 68.6	(l) 81.4	(m) 67.0	(n) 78.9	(o) 10.8	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	98.9	73.4	64.8	92.6	71.9	63.2	87.3	70.6	34.4	83.7	33.0	82.0	31.6	79.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.1	(b) 23.1	(c) 20.6	(e) 28.6	(f) 102.7	(g) 1.5	(h) 4.0	(i) 27.5	(j) 105.5	(k) 26.6	(l) 107.9	(m) 25.8	(n) 110.1	(o) 24.7	(p) 113.0		
(a) 26.1	(b) 23.1	(c) 20.6	(e) 28.2	(f) 74.3	(g) 1.6	(h) 3.4	(i) 27.0	(j) 76.8	(k) 26.1	(l) 78.7	(m) 25.2	(n) 80.6	(o) 24.1	(p) 83.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	67.6	67.2	63.7	58.0	52.7	48.6	47.6	49.0	52.1	55.7	60.0	63.6		
	DBStd	8.65	6.79	6.14	6.06	4.99	3.89	3.73	3.53	3.66	4.17	5.20	5.89	6.33		
	HDD50	272	0	0	0	3	16	70	90	63	22	7	1	0		
	HDD65	3272	49	42	99	216	380	492	539	497	387	293	173	105		
	CDD50	2861	545	482	425	243	101	28	16	31	84	183	302	421		
	CDD65	388	129	104	59	6	0	0	0	0	4	24	62			
	CDH74	4174	1314	931	553	89	2	0	0	0	13	140	372	760		
	CDH80	1696	607	379	197	17	0	0	0	0	1	34	128	333		
(14) Wind	WSAvg	8.6	9.6	9.0	8.4	7.2	7.6	6.9	8.1	8.7	9.3	9.2	9.6	9.8		
(15) Precipitation	PrecAvg	29.9	2.1	1.4	2.0	2.2	2.6	2.5	2.6	2.8	3.1	2.9	2.6	2.5		
	PrecMax	38.7	5.9	5.3	8.0	4.7	7.0	5.0	5.3	6.0	6.3	6.9	5.2	7.7		
	PrecMin	18.3	0.0	0.1	0.2	0.3	0.4	0.6	0.6	0.9	1.1	1.0	0.9	0.2		
	PrecStd	5.1	1.1	1.2	1.5	1.2	1.5	1.0	1.3	1.3	1.2	1.3	1.2	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.6	96.2	92.7	81.9	71.7	63.9	63.9	67.7	75.2	84.4	89.7	97.4		
		MCWB	68.8	67.2	65.6	61.4	57.9	54.4	52.5	55.8	59.2	62.9	66.3	68.0		
	2%	DB	92.8	89.7	85.4	75.9	66.8	60.1	60.0	63.2	69.3	77.6	83.5	89.0		
		MCWB	67.8	67.2	65.0	60.7	56.9	52.9	51.6	53.2	57.2	61.0	65.1	66.1		
	5%	DB	86.4	84.4	80.1	71.5	63.3	57.8	57.3	60.1	65.2	72.2	77.7	82.6		
		MCWB	67.1	66.8	64.1	59.7	55.4	52.0	50.5	51.6	55.6	59.3	63.0	64.7		
	10%	DB	80.8	79.4	75.2	67.5	60.6	55.8	55.0	57.2	61.4	67.2	72.4	76.5		
		MCWB	65.6	65.3	62.8	58.1	54.3	51.1	49.5	50.3	53.5	57.5	61.2	63.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.3	72.4	69.5	65.0	60.5	57.2	54.9	57.8	61.8	66.1	69.7	70.4		
		MCDB	88.5	86.4	81.9	74.4	66.7	59.6	59.5	64.6	70.2	77.5	83.9	85.5		
	2%	WB	70.6	69.9	67.2	62.6	58.6	54.9	52.8	54.6	58.7	62.6	66.3	68.0		
		MCDB	84.6	82.1	78.9	70.6	63.5	58.3	57.5	60.7	66.7	73.9	79.8	82.1		
	5%	WB	68.7	68.2	65.5	61.0	56.8	53.0	51.3	52.5	56.2	60.2	64.2	66.2		
		MCDB	82.0	79.2	75.8	68.4	61.8	56.5	55.7	58.1	63.5	69.4	74.6	78.9		
	10%	WB	66.7	66.4	63.8	59.3	55.1	51.6	50.2	51.0	54.3	58.2	62.0	64.1		
		MCDB	78.8	76.6	73.2	66.0	59.8	55.0	54.2	55.9	60.3	66.6	70.9	74.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.1	21.8	20.7	19.3	16.4	15.3	15.5	16.8	18.2	20.4	21.0	22.2		
		MCDBR	33.8	31.4	29.6	26.9	21.9	18.4	20.0	23.0	26.5	32.2	32.9	34.3		
		MCWBR	12.0	11.5	11.2	12.6	13.0	12.3	12.9	14.0	15.5	17.0	15.6	14.1		
	5% WB	MCDBR	27.7	25.4	24.4	21.8	19.5	16.4	16.7	20.6	23.5	27.3	27.4	28.1		
		MCWBR	11.4	10.7	10.4	11.0	12.4	11.7	11.6	13.4	14.5	15.4	14.2	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.375	0.368	0.345	0.334	0.318	0.302	0.317	0.313	0.347	0.346	0.351	0.363		
	taud		2.421	2.461	2.517	2.527	2.550	2.589	2.524	2.536	2.433	2.461	2.473	2.448		
	Ebn at Noon		304	298	292	275	261	257	257	279	286	302	309	309		
	Edh at Noon		39	36	31	27	23	20	23	26	33	35	36	38		
(44) All-Sky Solar Radiation	RadAvg		2181	1872	1502	1058	713	569	638	893	1258	1662	1947	2173		
	RadStd		148	117	97	62	39	43	34	38	91	124	121	132		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	+0.90	N/A	N/A	+1.42	N/A	N/A	-32	-234	+308	+110

Nomenclature: See separate page